

AU200 RESEARCH ARCHIVE

Complete Technical Record — All Sessions

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Trade Cartel | Repository: kushverma7/Trade_Cartel

PURPOSE: This document is a machine-readable archive of every AU200 research session, strategy, test, and finding. It is the source of truth for the state of AU200 knowledge. A downstream AI or human must be able to reconstruct the full picture from this document alone without access to any other file or conversation.

■ **NO AU200 RESULT HAS BEEN VALIDATED. No row exists in RESULTS_LEDGER.md. All numbers below are either (a) unverified screenshot claims, (b) simulation results produced with BUG-019 present, or (c) Python backtest results on real data run 2026-08-05. Nothing is cleared for live trading.**

SECTION 0 — EXECUTIVE STATE OF AU200 RESEARCH

Current Best Candidate

NONE. No AU200 strategy has survived validation. The single most-tested candidate (UT Bot + EMA200, session 10:00–12:00 AEST) returned PF 0.83 on 6 years of real 5-minute data with BUG-019 corrected. A 576-combination parameter grid found zero configurations with PF > 1.0. Zero AU200 rows exist in RESULTS_LEDGER.md.

What Has Been Firmly Ruled Out

Approach	Test	Ruling Metric	Status
UT Bot + EMA200, 10:00–12:00 AEST	Python backtest 2020–2026, BUG-019 fixed	PF 0.83, losing all 7 years	KILLED
Gap direction (open vs prev close)	Forward-return t-test on real data	t = −0.17 / −0.33 (not significant)	KILLED
Opening range breakout (first 15min)	Forward-return t-test on real data	t = −0.16, win% = 49%	KILLED
Opening bar momentum (follow 10:00 bar direction)	Forward-return t-test on real data	t = −0.20, win% = 48.6%	KILLED
576 UT Bot parameter combos (SL/TP/trail/ATR multiplier)	Python grid scan	Zero PF > 1.0	KILLED
London open strategy (15:00 AEST, SL=10, TP=15)	Python backtest	PF 0.47, −3,068 pts net	KILLED
Screenshot 67–69% WR results (9-row table)	Code review + BUG-019 identification	BUG-019 in every file examined	UNTRUSTWORTHY

What Remains Open

Open Question	Why Not Answered	Priority
Does any signal have edge in AU200 10:00–12:00 AEST? Other signals (MACD, RSI divergence, volume breakout) not tested.	Not tested.	HIGH
Does the 15:00 AEST (edge on 3.77) edge +1.97 pts/hold over 60min SL backtest PF 0.47. A volatility-adaptive exit (trailing only, no hard SL) was not tested.	Not tested.	HIGH
What is the AU200 true mintick?	Assumed 0.1 from data distribution. Not confirmed off TradingView charts.	MEDIUM
Does the AU200 have edge on 15m or 30m trading? 5m is minute only. XAUUSD improved on 30m; AU200 not tested.	Not tested.	MEDIUM
Can Kronos directional accuracy validate a tick fence?	Kronos 2 rights blocked (HuggingFace 403 at proxy). Cannot run.	BLOCKED

One-Paragraph Honest Status

As of 2026-08-05, AU200 has NO validated strategy. The session-based research that produced the nine-row screenshot table (showing PF 4.07–7.11) has been found to contain BUG-019 (excursion update before stop check) in every file examined. Correcting BUG-019 and re-running on six years of real 5-minute data yields PF 0.83. A 576-combination parameter grid finds zero profitable configurations. The 10:00–12:00 AEST session has no statistically significant directional edge (t-stats all below 2.0). A mild edge exists at 15:00 AEST (t=3.77, +1.97 pts/hold over 60min) but fixed-SL strategies cannot capture it profitably. The user's target of 50 pts/week consistent is not achievable from any tested configuration at 1 contract. No AU200 row exists in RESULTS_LEDGER.md. The next required step is to test non-fixed-SL approaches (trailing stop only) at 15:00 AEST and to scan signals other than UT Bot in the morning session.

SECTION 1 — DATA & MARKET SPECIFICATION

Instrument Names Used

Context	Symbol Used	Source
Pine strategy files	AUS200 CFD (described in headers)	Not specified
Python backtest (2026-08-05)	CAPITALCOM:AU200 (5m CSV)	User-supplied upload
TradingView attempt	CAPITALCOM:AU200	MCP call failed (CDP error)

Data File (Python Backtests)

Field	Value
Filename	bc3182e6-au200_aud_5m.csv
Path	/root/.claude/uploads/7c610517-7865-597e-b767-c39eacd908c6/bc3182e6-au200_aud_5m.csv
Rows	139,898
Date range	2020-08-05 00:00 UTC to 2026-08-04 00:00 UTC
Columns	timestamp (UTC), open, high, low, close, volume
Timeframe	5-minute bars
Currency	AUD (prices in index points, e.g. 6,024.0)
Price range	5,760.2 to 9,240.5 index points
Mintick (confirmed)	0.1 points (inferred from smallest observed close diff; not read off TradingView chart)
Volume	Populated; not used in any tested strategy

Hours 16–21 AEST have sparse data gaps before 2023, causing forward-return artifacts for those hours. UTC hours 0, 1, 23 carry 10:00–12:00 AEST data (UTC+10 and UTC+11).

Session & Trading Hours

Parameter	Value	Notes
Target session (all Pine files)	10:00–12:00 AEST	Australia/Sydney timezone string (DST-safe)
DST offset (AEST)	UTC+10	Standard; clocks do NOT change for AEST itself
DST offset (AEDT)	UTC+11	Oct–Apr; AU clocks forward
10:00 AEST in UTC	00:00 UTC	During AEST (winter)
10:00 AEDT in UTC	23:00 UTC previous day	During AEDT (summer)
Session bars in data	36,320 of 139,898 total (25.9%)	Correctly filtered in Python
DST bug (earlier sessions)	hour(time, 'UTC+10') used	Fired 1hr early Oct–Apr. Fixed to Australia/Sydney.
London open session (tested)	15:00–16:00 AEST	05:00–06:00 UTC; used for edge scan only

Contract Specification

Parameter	Value	Source
Point value	\$100 AUD per point	Screenshot report (UNVERIFIED)
Mintick	0.1 pt (inferred) — VERIFY on chart	Python data analysis
Margin	NOT STATED IN SESSION	N/A

Broker	NOT STATED IN SESSION	User trades CFD; broker unspecified
Slippage (Pine files)	0 in au200_tbt_flip; 1 tick elsewhere	CODE
Slippage (Python tests)	1.0 pt per side	Assumption; mintick=0.1 → 10 ticks
Commission (Pine — base flip)	\$1.00 AUD per contract	cash_per_contract
Commission (Pine — tbt flip)	\$0.50 per order	cash_per_order (both sides combined)
Commission (Python tests)	\$1.00 AUD per side (\$2.00 round-trip)	Assumption
Starting equity (all tests)	\$50,000 AUD	CODE

Differences from Gold (XAUUSD) Observed

Dimension	XAUUSD (Gold)	AU200 (ASX 200 CFD)
Trading hours	24h Sunday–Friday	Session-anchored, primary 10:00–16:00 AEST
Session edge	London + NY open hours show strong bias	10:00–12:00 AEST has NO significant edge
Bars/week in champion	~2.3 trades/week	2.52 trades/day claimed (screenshot)
Cost sensitivity	Break-even at ~1.2–1.5 pts slippage	NOT MEASURED (no profitable baseline)
Gap behaviour	Continuous; gaps only on Sunday open	Daily gaps at 10:00 AEST open
Trend filter	EMA/SMA trend filters hurt results (3 tests)	NOT TESTED
ATR environment	~\$1–\$5 moves per 5m bar typical	~15–50 pt moves per 5m (assumed from SL choices)

SECTION 2 — STRATEGIES & VARIANTS TESTED

2.1 AU200 Base + Flip Strategy (au200_base_flip_v1.pine)

Field	Value
Session chat	Session chat + code review. File was read in full before context compaction. File LOST to container reclamation (push 403 failure). Not in re
Source label	SESSION CHAT
Status	REJECTED — no backtest result; code had BUG-019 in exit block; file lost

Entry Rules (complete specification):

Parameter	Rule	Default
Timeframe	5-minute chart	Fixed
Asset	AUS200 CFD	Fixed
Session window	9:50, 9:55, or 10:00 AEST (user toggle per bar)	All three ON
EMA filter	close > EMA(200) for long; close < EMA(200) for short	ON
RSI filter	RSI(14) > 50 for long; RSI(14) < 50 for short	ON
SuperTrend filter	SuperTrend(3.1, 97) bullish for long; bearish for short	ON
State gate	position_size == 0 AND _dir == 0 (flat only)	Fixed
Logic	ALL three filters must agree with direction	AND gate

Exit Rules:

Exit Type	Rule	Default
Hard SL	Flat price below entry (long) or above entry (short)	15 pts
Trail distance	Trail stop from excursion peak/trough	30 pts
Trail trigger	Trail arms once price moves this far in favour	5 pts
TP1	Close 50% of position at fixed target	20 pts
TP1 breakeven	Move SL to entry price after TP1 hit	ON (toggle)
EOD	Force-close all positions at 12:00 AEST	Fixed
Flip on SL	Reverse direction on SL hit	ON
Flip SL multiplier	Flip stop = SL × this multiplier (so 30 pts on flip)	2.0
Max flips	Maximum reversals per original entry	3

Cost Model:

Parameter	Value
commission_type	strategy.commission.cash_per_contract
commission_value	\$1.00 AUD
currency	AUD
initial_capital	\$50,000
process_orders_on_close	true
slippage	NOT STATED IN SESSION (parameter not confirmed in read)

BUG-019 PRESENT: In the exit block, `v_pk := math.max(v_pk, high)` was confirmed to appear BEFORE if `low <= sl_p` on the same bar. Win rate is inflated.

RESULT: NOT MEASURED. No backtest was ever run on this strategy. The user stated 'this is the most profitable strategy yet' after seeing signals on the chart — that is a visual impression, not a backtest result.

FILE STATUS: LOST. Container was reclaimed before git push succeeded.

2.2 AU200 TBT + Flip System (au200_tbt_flip_backtest.pine)

Field	Value
Provenance	Session chat; code was pasted by user and read in full. File LOST to container reclamation.
Source label	SESSION CHAT
Status	REJECTED — BUG-019 confirmed in exit block; slippage=0; claimed result is unverified Pine comment only

Entry Rules:

Parameter	Rule	Default
Timeframe	5-minute chart	Fixed
Entry bar	Exactly hour==10 and minute==0 (10:00 AEST)	Fixed
Gap direction	open - close[1] > 0 → long; < 0 → short	Primary trigger
SuperTrend filter	SuperTrend(10, 1.5) must agree with gap direction	ON
OLS trendline direction filter	OLS trendline direction must agree (pivot-high resistance descending for long; pivot-low support ascending for short; close crosses trendline)	ON

Exit Rules:

Exit Type	Rule	Default
Hard SL	5 pts per leg (very tight)	5 pts
Trail	75 pts from excursion peak, arms at +5 pts	75 pts
TP1 signal	Visual label only at +20 pts — NOT a strategy exit	20 pts
Max flips	Up to 6 reversals on SL hit	6
Flip SL	Same 5 pts on each flip	5 pts

Cost Model:

Parameter	Value
commission_type	strategy.commission.cash_per_order
commission_value	\$0.50 (both sides per order)
slippage	0 (EXPLICITLY ZERO — unrealistic)
initial_capital	NOT STATED IN SESSION

Claimed Result (Pine file header comment — NOT validated):

Claimed Metric	Value	Validation Status
Profit Factor	2.10	NOT VALIDATED — Pine comment only
Win Rate	9.7%	NOT VALIDATED
Years	8/8 (2019–2026)	NOT VALIDATED

Net points	11,819 pts	NOT VALIDATED
Trade count	NOT STATED IN SESSION	NOT MEASURED
Max Drawdown	NOT STATED IN SESSION	NOT MEASURED
Commission in claim	NOT STATED IN SESSION	NOT MEASURED
Slippage in claim	0 (explicit in code)	UNREALISTIC

BUG-019 CONFIRMED: $v_pk := \text{math.max}(v_pk, \text{high})$ appears BEFORE if $\text{low} \leq \text{sl_p}$ in the strategy's position management block. On bars where both conditions fire, excursion is updated to the bar's high before the stop resolution — inflating win rate.

RULING: The claimed PF 2.10 cannot be used as a result. It is a comment in the Pine file with no trade count, no drawdown, no cost model confirmation, and no DSR/PBO. The code has BUG-019 and slippage=0. Discard the number entirely.

2.3 Top & Bottom Entry Indicator (top_bottom_entry_v1.pine)

Field	Value
Provenance	Session chat; built from user-supplied non-repainting trendline code.
Type	INDICATOR — no strategy.entry, no backtest possible without conversion
File status	LOST (container reclamation before push)
Backtest result	NOT MEASURED

Mechanism:

Parameter	Rule	Default
Lookback	OLS linear regression over N bars on close[1] (non-repainting)	9 bars
Resistance line	Highest pivot above OLS line; slope optimised (30-step binary search)	—
Support line	Lowest pivot below OLS line; slope optimised (30-step binary search)	—
BOT signal	$\text{close} > \text{resistance_line} + \text{atr_mult} \times \text{ATR14}$	—
TOP signal	$\text{close} < \text{support_line} - \text{atr_mult} \times \text{ATR14}$	—
ATR threshold	close must exceed line by at least this fraction of ATR14	$0.3 \times \text{ATR14}$
Cooldown	Minimum bars between signals	3 bars
Alternating	Must flip direction each signal (no consecutive BOT/BOT)	ON

2.4 Top & Bottom Strategy (top_bottom_strategy_v1.pine)

Field	Value
Provenance	Session chat; built from top_bottom_entry_v1 + exit logic.
File status	LOST (container reclamation before push)
Backtest result	NOT MEASURED

Exit Rules (beyond trendline indicator):

Exit Type	Value
Hard SL	20 pts
Trail	15 pts from excursion peak, arms at +5 pts

Reverse exit	Close and reverse on opposite trendline signal (use_rev_exit=true)
Commission	\$1 AUD per contract, cash_per_contract
Initial capital	\$50,000

BUG-019 PROBABLE: the trail management block likely updates excursion before stop check (same pattern as the other files — same developer session, same exit template).

2.5 AU200 UT Bot Session Strategy v1 (au200_utbot_session_v1.pine)

Field	Value
Provenance	Built 2026-08-05 in this session; pushed to repo.
File path	research/au200_utbot_session_v1.pine
Status	TESTED (Python backtest); REJECTED — PF 0.83
BUG-019	FIXED (excursion update confirmed AFTER stop check)

Entry Rules:

Parameter	Rule	Default
Entry signal	UT Bot direction change (ATR trailing stop crossover)	ATR(10), mult=1.5
EMA filter	close > EMA(200) for long; close < EMA(200) for short	ON (toggle)
Session	10:00–12:00 AEST via Australia/Sydney timezone	Fixed
State gate	posDir == 0 (flat) and not just_stopped	Fixed

Exit Rules:

Exit Type	Rule	Default
Hard SL	Fixed distance from entry	15 pts
Trail	Trail from excursion ref	30 pts
Trail trigger	Arms after price moves this far in favour	5 pts
TP1	Close full position at fixed target (partial not implemented)	20 pts
TP1 breakeven	Set SL = entry after TP1	ON
EOD	Hard close at 12:00 AEST	Fixed
Flip	Reverse on SL hit	SL × 2.0, max 3 flips/session

Backtest Result (Python, 2026-08-05, BUG-019 FIXED):

Metric	Value
Period	2020-08-11 to 2026-08-03
Total trades	2,727
Win Rate	37.1%
Profit Factor	0.83
Average winner	+14.0 pts
Average loser	−10.0 pts
Net points	−2,997.1
Net AUD (after commission)	−\$8,451

Max Drawdown	−17.0%
Weekly median (pts)	−10.0
Weekly mean (pts)	−9.6
Weekly std	42.9 pts
% weeks positive	38%
Best week	+135.5 pts
Worst week	−126.6 pts
Slippage modelled	1.0 pt per side
Commission	\$1.00 AUD per side

Year-by-Year Breakdown:

Year	Trades	Win Rate	Net Pts
2020	179	37%	−404
2021	556	39%	−395
2022	538	38%	−263
2023	340	37%	−371
2024	359	36%	−577
2025	408	36%	−639
2026	347	36%	−347

Exit Breakdown:

Exit Type	Count	Net Pts	Win Rate
SL (stop-out)	1,134	−12,693	0.7% (mostly round-trip to 0)
TP1	550	+10,450	100%
EOD (hard close)	1,043	−754	43.6%

VERDICT: REJECTED. PF 0.83. Losing every year. The stop-out block is the primary drag (−12,693 pts net). The flip engine is not helping — SL hit → flip → SL hit chains are occurring. No parameter combination in 576-combo grid produced PF > 1.0.

2.6 London Open Strategy (15:00 AEST, Python-only)

Field	Value
Provenance	Python-only test run 2026-08-05; no Pine file written
Entry	15:05 AEST bar; EMA50 direction filter
SL	10 pts
TP	15 pts
Exit	By 16:00 AEST if neither SL nor TP hit
Status	REJECTED — PF 0.47

Metric	Value
Period	2020-08-05 to 2026-08-03
Total trades	713
Win Rate	26.6%
Profit Factor	0.47
Net pts	−2,642.3
Net AUD	−\$4,068
Slippage	1.0 pt per side
Commission	\$1.00 AUD per side
% weeks positive	22%

VERDICT: REJECTED. The 15:00 AEST raw session edge (+1.97 pts avg over 60-min hold, $t=3.77$) is real but a fixed-SL strategy cannot capture it — the SL is hit 66% of the time before the move develops.

SECTION 2.7 — SCREENSHOT NINE-ROW TABLE (UNVERIFIED)

Source: Screenshot from a separate 'Research claims verification' session. Files from that session were LOST. Numbers cannot be reproduced. No drawdown, no cost model, no trade count year-by-year. BUG-019 was identified in every Pine file that could be inspected. These numbers are archived for completeness ONLY and must not be acted on.

Strategy	N trades	Win Rate	PF	Net 2019–2026	T/day	Slippage	Commission
UT Bot Session 10:00–12:00	4,893	67.7%	4.07	\$5,142,420	2.52	NOT STATED	NOT STATED
UT Bot + EMA Hybrid v2	4,407	74.4%	7.11	\$5,820,000 *	2.27	NOT STATED	NOT STATED
UT Bot OR Breakout (10:00–10:10)	2,630	67.9%	4.07	\$2,696,410	1.36	NOT STATED	NOT STATED
ST+RSI Trail30 EMA filtered	3,001	67.4%	3.46	\$2,938,810	1.55	NOT STATED	NOT STATED
ST only Trail30 no filter	6,298	67.9%	3.65	\$6,725,280	3.25	NOT STATED	NOT STATED
EMA Trail system (best exit V1)	2,752	39.6%	1.06	\$68,336	NOT STATED	NOT STATED	NOT STATED
UT Bot + ST filter	3,797	67.0%	3.98	\$3,858,950	1.96	NOT STATED	NOT STATED
UT Bot + ST+RSI filter	2,430	67.1%	4.02	\$2,483,050	1.25	NOT STATED	NOT STATED
EMA zone filter F6 (EMA50+200)	3,034	68.7%	4.39	\$3,309,180	NOT STATED	NOT STATED	NOT STATED

* asterisk: described as 'hybrid estimate from Python backtest engine'

Missing from every row:

Required Field	Status
Max Drawdown	NOT MEASURED
Starting equity	NOT STATED IN SESSION
Slippage (ticks or points)	NOT STATED IN SESSION
Commission per side	NOT STATED IN SESSION
Fixed lots vs compounding	NOT STATED IN SESSION
Year-by-year breakdown	NOT MEASURED
Out-of-sample split	NOT MEASURED
DSR statistic	NOT MEASURED
PBO statistic	NOT MEASURED
Source (TradingView live vs Python)	Only row 2 labelled; rest unknown
BUG-019 status	CONFIRMED in at least 2 of the underlying files

WIN RATE CLUSTERING ANALYSIS: Seven of nine strategies show 67.0–68.7% WR despite different entry logic. This is the signature of a shared exit bug (BUG-019). A trailing-stop trend-following system with realistic cost and correct code should show 35–45% WR. The only honest-shaped row is EMA Trail at 39.6% WR / PF 1.06 — the single row that either does not use the buggy exit block, or has correct operation order. No number from the 67–69% cluster should be treated as real until the exit block is corrected and the backtest re-run.

SECTION 3 — CROSS-CUTTING FINDINGS

3.1 Session Time-of-Day Edge Scan

Run 2026-08-05 on real data. Method: buy at start of each AEST hour, hold 60 min, measure average return and t-stat. No signal, no filter — raw hourly directional bias.

Hour (AEST)	N bars	Avg return (pts)	T-stat	Win %	Significance
00:00	2,086	−0.91	−2.09	45.7%	** SHORT BIAS
01:00	2,074	+0.70	1.69	50.7%	
02:00	2,078	−0.17	−0.41	49.0%	
03:00	2,055	−0.76	−2.44	46.9%	** SHORT BIAS
04:00	2,060	−0.68	−2.14	49.5%	** SHORT BIAS
05:00	2,053	+0.77	+2.39	49.1%	** LONG BIAS
06:00	2,025	−3.13	−4.43	44.1%	**** STRONG SHORT
07:00	787	−2.08	−2.25	48.2%	** SHORT BIAS
09:00	346	−2.96	−2.06	44.5%	** SHORT BIAS
10:00 (SESSION)	18,149	−0.21	−1.48	48.6%	NOT SIGNIFICANT
11:00 (SESSION)	18,170	+0.26	+2.19	49.7%	* MILD LONG
12:00	18,175	+0.17	+1.83	49.5%	
13:00	18,183	+0.15	+1.69	49.0%	
14:00	18,135	+0.33	+3.84	49.9%	*** LONG BIAS
15:00 (LONDON)	18,094	+1.97	+3.77	54.4%	*** LONG BIAS
16:00	1,031	+20.79*	+3.02*	52.6%*	ARTIFACT (data gaps)
17:00	1,734	+13.12*	+2.06*	53.5%*	ARTIFACT (data gaps)
18:00	2,124	+1.04	+2.52	52.7%	** LONG BIAS
21:00	2,083	+10.07*	+4.55*	51.3%*	ARTIFACT (data gaps)

* Hours 16, 17, 21 have data gaps in 2020–2022 causing the shift(-12) forward return to span across multi-day breaks. These t-stats are data artifacts, not real edge.

3.2 What Transferred from Gold Research

Finding from Gold (XAUUSD)	Applied to AU200?	Result
Session filter adds edge	YES — 10:00–12:00 AEST	No edge found (t<2)
Trailing stop beats fixed TP	YES — trail in all AU200 files	NOT TESTED in isolation
Trend filters hurt (3 independent tests)	NOT TESTED on AU200	N/A
30m chart beats 5m	NOT TESTED on AU200	N/A
Pyramiding adds return	NOT IMPLEMENTED on AU200	N/A
BUG-019 inflates results	YES — identified in all AU200 files	Confirmed effect
BUG-021 (slippage in ticks)	YES — applied to AU200 cost model	slippage=1 → 1pt/side
Corrected null required before VALID	NOT RUN on AU200	Required before VALID

3.3 Microstructure Observations

Observation	Source
AU200 CFD data has full 24h coverage but sparse data 16:00–22:00 AEST before 2023	Python data analysis
Mintick is approximately 0.1 pts (not 1.0 as originally assumed)	Python data analysis
The 10:00 AEST opening bar has NO momentum follow-through ($t=-0.20$)	Forward-return scan
The 15:00 AEST hour has a genuine long bias ($t=3.77$, repeatable 5/6 years)	Hour scan
2022 was the exception year for the 15:00 AEST edge (avg -2.71 , $t=-2.09$)	Year-by-year scan
\$100/pt point value reported in screenshot; not confirmed in any code	SCREENSHOT (unverified)

SECTION 4 — FAILED FAMILIES (KILL LIST)

Every approach tested and rejected on AU200, with the ruling metric. These must not be re-tested without a specific reason why the next attempt differs materially from the rejected one.

Family	Specific Form	Ruling Metric	Kill Reason
UT Bar Entry	Trailing stop crossover (ATR 10, mult 1.5) + EMA200 filter	28000 pts net edge in signal years	EMA filter makes it worse.
OTC Comparable	TP 15–30, trail 20–40, ATR mult 1.0–2.0, filter 0/B, EMA cutoff	57600 pts	Exhaustive search found no configuration that works
Gap direction	10:00 AEST open vs previous close, hold session t = −0.17 / −0.33		Coin flip. No statistical edge in 6 years.
Opening range break	5 min high/low; enter on breakout, hold to 12:00	0.16, win% = 49%	Coin flip. No edge in 6 years.
Opening bar momentum	Follow direction of 10:00 AEST bar, hold 30 min	−0.20, win% = 48%	Strong open filter (bar > 1 ATR) makes no difference.
London open	15:05 AEST entry, EMA50 filter, SL=10, TP=15, BR by 1.52 pts	71600 pts, \$4,000 net	Raw session edge (+1.97 pts avg) cannot survive a fixed
All screenshot strategies (67–69% WR) from prior session		BUG-019 confirmed in every respect	UG-019 confirmed in every respect
EMA trail	(39.6% WR) from screenshot; exact rules unknown	PPE 0.06	is not useful (near breakeven). Rules not recoverable (file lost).

SECTION 5 — OPEN QUESTIONS & RECOMMENDED NEXT TESTS

Priority	Test	Hypothesis	Why Unblocked
1 — HIGH	Check SL500145 Strategy API 15:00 AEST (7:30 AM UTC) for 5/6 years) might be captured by DataLink table, requires new Python script (but before the morning session)		
2 — HIGH	Check Polaris signals at 14:00-15:00 AEST (10:00-11:00 AM UTC) for session, volume breakout, VWAP deviation, requires software (bloomberg or high)		
3 — HIGH	Confirm AU200 mintick off Caching, assumed 0.1 from data. If 0.5 requires slippage open trading view (CDP connection failed from cloud)		
4 — MED	5m and 30m charts on XESTUSD using signal on 30m chart. AU200 5m may have to build 15m/30m data file from user		
5 — MED	Corrected null on any signal, verify if any result enters RESULTS_LEDGER (BU6023 protocol) a candidate exists		
6 — MED	Ar-by-year 2022 gas analysis on 15:00 AEST (11:00 AM UTC) bias at 15:00). Is it linked to macro regime (Data available, is 30 min)?		
7 — LOW	Run Kronos validate on 19:50 AEST (15:00 UTC) for fractional accuracy = no BLOCKED though, error of 400 and container proxy. Needs weight files.		

SECTION 6 — BUG REGISTRY (AU200-RELEVANT ENTRIES)

Bug ID	Name	Severity	Effect on AU200 Results
BUG-019	Internal block size not being checked on save	CRITICAL	Confirmed in au200_tbt_flip_backtest.pine and au200_base_flip_v1.pine. Inflates win
BUG-020	Slippage=0 Pine 200 page flip (explicit) slippage variable known in base_flip file. Any run with wrong slippage is overstated. Correct: read mintick	CRITICAL	
BUG-021	Edit symmetry — long/short m — any block applied to long block misses the short block. All AU200 files have symmetric long/short exit stru	HIGH	
BUG-022	Confounded null — measure AU200 nothing has been run yet. When one is, it must use the same filter set as the strategy, varying only the e	HIGH	
BUG-023	DST bug (unrelated), 'UTC+10' fires 1hr early Oct 2019	CRITICAL	All AU200 files originally used UTC+10 hardcoded. Fixed to Australia/Sydney before files were lost.

SECTION 7 — FILE & SOURCE INDEX

Files Present in Repo (2026-08-05)

File path	Type	Status	Description
research/au200_utbot_session_v1.pine	Pine Strategy	COMMITTED	UTBOT + session filter; BUG-019 fixed; pine_lint CLEAN; issue_count=0
research/au200_utbot_backtest.py	Python backtest	COMMITTED	BUG-019-compliant backtest. Produces the 2727-trade result above.
research/build_au200_archive.py	PDF generator	COMMITTED	Generates this document
research/AU200_RESEARCH_ARCHIVE.pdf	PDF archive	COMMITTED	This document
trader_playbooks/AU200_KNOWLEDGE_BASE.md	Knowledge doc	COMMITTED	Comprehensive AU200 knowledge base, 333 lines, all source-tagged

Files Mentioned in Session but NOT in Repo (LOST)

File	Why Lost	Content Recovered?
indicators/au200_indicator_v1.pine	Container reclamation before push	PARTIAL — from summary
strategies/au200_base_flip_v1.pine	Container reclamation before push	FULL — from session chat
research/au200_tbt_flip_backtest.pine	Container reclamation before push	FULL — from session chat
indicators/top_bottom_entry_v1.pine	Container reclamation before push	PARTIAL — from summary
strategies/top_bottom_strategy_v1.pine	Container reclamation before push	PARTIAL — from summary
research/au200_utbot_ema_hybrid_v2.pine	Not in current session repo	REFERENCED only (PR title)

External Data

File	Location	Status
bc3182e6-au200_archive.zip	uploads/7c610517-7865-597e-b767-c39eac408a6/bc3182e6-au200_archive.zip	Will be lost on container reclaim

Sources Used to Build This Document

Source	Content	Trust Level
Repo: trader_playbooks/AU200_KNOWLEDGE_BASE.md	All prior session findings, tagged	HIGH
Repo: research/au200_utbot_backtest.py	Actual Python results	HIGH
Repo: research/au200_utbot_session_v1.pine	Pine code	HIGH
Repo: MEMORY.md	Session history, validated findings	HIGH
Repo: trader_playbooks/bugs/BUG_REGISTRY.md	Bug descriptions	HIGH
Session chat (compacted summary)	Pine scripts, table, strategy specs	MEDIUM (chat, not verified code)
Screenshot (nine-row table)	Unverified performance numbers	LOW — do not act on

SECTION 8 — COMPLETE PINE SCRIPT CODE

Only code that exists in the repo is reproduced here in full. Code that was lost (au200_base_flip_v1, au200_tbt_flip_backtest, top_bottom files) is summarised in Section 2. The full source of those files was in the session chat and is recoverable from the transcript at: /root/.claude/projects/-home-user-Trade-Cartel/7c610517-7865-597e-b767-c39eacd908c6.jsonl

8.1 au200_utbot_session_v1.pine (complete)

```
// =====  
// AU200 UT Bot Session Strategy v1.0  
// =====  
// SPEC (Phase 0):  
// Asset: AU200 CFD | TF: 5-minute recommended  
// Entry: UT Bot buy/sell signal (ATR trailing stop crossover) + EMA200 direction  
// Only within session 10:00-12:00 AEST (Australia/Sydney)  
// Exit: SL=15pts | Trail=30pts (arms at +5pts from entry) | TP1=20pts (50%)  
// Flip on SL hit: SL×2.0, max 3 flips | Hard close at 12:00 AEST  
// Costs: commission_type=cash_per_contract, $1 AUD | slippage=1 tick (1pt)  
// NOT IN SCOPE: news filter, volume profile, multi-TF  
//  
// BUG-019 FIX: excursion peak (v_pk) updated AFTER stop check on every bar.  
// Previously updating v_pk before checking sl → inflated win rate.  
//  
// MINTICK ASSUMPTION: syminfo.mintick assumed = 1.0 (AU200 CFD standard).  
// Verify on chart before live use. slippage=1 means 1 tick = 1pt.  
// =====  
  
//@version=6  
strategy(  
    title          = "AU200 UT Bot Session v1",  
    overlay        = true,  
    initial_capital = 50000,  
    default_qty_type = strategy.fixed,  
    default_qty_value = 1,  
    commission_type = strategy.commission.cash_per_contract,  
    commission_value = 1.0,  
    slippage        = 1,  
    process_orders_on_close = true,  
    max_labels_count   = 500,  
    max_lines_count    = 500  
)  
  
// ■■ Inputs ████████████████████████████████████████████████████████████████  
grpUT      = "UT Bot"  
i_atrLen    = input.int(10,     "ATR Length",           minval=1, group=grpUT)  
i_atrMult   = input.float(1.5,  "ATR Multiplier",       minval=0.1, step=0.1, group=grpUT)  
  
grpEMA      = "EMA Filter"  
i_emaLen    = input.int(200,    "EMA Length",         minval=10, group=grpEMA)  
i_useEma    = input.bool(true,  "Require EMA filter",  group=grpEMA)  
  
grpSess      = "Session"  
i_sessStart = input.int(10,     "Session start hour (AEST)", minval=0, maxval=23, group=grpSess)  
i_sessEnd   = input.int(12,     "Session end hour (AEST)",   minval=0, maxval=23, group=grpSess)  
i_tz        = input.string("Australia/Sydney", "Timezone", group=grpSess)  
  
grpExit      = "Exit"  
i_sl        = input.float(15.0, "Stop Loss (pts)",        minval=1.0, group=grpExit)  
i_tp1       = input.float(20.0, "TP1 (pts)",            minval=1.0, group=grpExit)  
i_tp1Pct    = input.float(0.50, "TP1 close fraction", minval=0.1, maxval=1.0, step=0.05, group=grpExit)  
i_useTP1BE  = input.bool(true,  "Move SL to BE after TP1", group=grpExit)  
i_trailPts  = input.float(30.0, "Trail distance (pts)",    minval=1.0, group=grpExit)
```

[illegible]

```

bool justFlipped = false

if posDir != 0
    bool isLong = posDir == 1

    // 1. CHECK STOP (must be before peak update – BUG-019 fix)
    bool stopHit = isLong ? low <= sl_p : high >= sl_p

    if stopHit
        // Flip logic
        if inFlip and flipCount < i_maxFlips
            // Open reverse position at SL price
            float flipSL = sl_p // entry is at current SL price
            float newSL = isLong ? flipSL - i_sl * i_flipSLMult : flipSL + i_sl * i_flipSLMult
            float newTP1 = isLong ? flipSL - i_tp1 : flipSL + i_tp1

            if isLong
                strategy.close("Long", comment="SL→Flip")
                strategy.entry("Short", strategy.short, comment="Flip↓")
            else
                strategy.close("Short", comment="SL→Flip")
                strategy.entry("Long", strategy.long, comment="Flip↑")

            posDir := -posDir
            entryPrice := flipSL
            sl_p := newSL
            tp1_p := newTP1
            trailRef := flipSL
            tp1Done := false
            trailArmed := false
            flipCount += 1
            justFlipped := true
        else
            // Close flat
            if isLong
                strategy.close("Long", comment="SL")
            else
                strategy.close("Short", comment="SL")
            posDir := 0
            entryPrice := na
            sl_p := na
            tp1_p := na
            trailRef := na
            tp1Done := false
            trailArmed := false
            inFlip := false
            justStopped := true

    // 2. Trail stop (armed check, AFTER stop already handled)
    if posDir != 0 // still in position (not just stopped)
        bool isLong2 = posDir == 1
        // Arm trail once price moves i_trailTrig pts in our favour
        if not trailArmed
            bool trigLong = isLong2 and high >= entryPrice + i_trailTrig
            bool trigShort = not isLong2 and low <= entryPrice - i_trailTrig
            if trigLong or trigShort
                trailArmed := true
                trailRef := isLong2 ? high : low

    // 3. UPDATE excursion peak (BUG-019 fix: AFTER stop check)
    if trailArmed
        if isLong2
            trailRef := math.max(trailRef, high)
            float newSL = trailRef - i_trailPts
            if newSL > sl_p
                sl_p := newSL

```


[illegible]

9.1 au200_utbot_backtest.py (complete)

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[illegible]

[illegible]

```

        if is_long2 and h >= entry_price + TRAIL_TRIG:
            trail_armed = True
            trail_ref = h
        elif not is_long2 and l <= entry_price - TRAIL_TRIG:
            trail_armed = True
            trail_ref = l

    if trail_armed:
        if is_long2:
            trail_ref = max(trail_ref, h)
            new_sl = trail_ref - TRAIL_PTS
            if new_sl > sl_p:
                sl_p = new_sl
        else:
            trail_ref = min(trail_ref, l)
            new_sl = trail_ref + TRAIL_PTS
            if new_sl < sl_p:
                sl_p = new_sl

# 3. TP1
if not tp1_done:
    tp1_hit = (pos_dir == 1 and h >= tp1_p) or (pos_dir == -1 and l <= tp1_p)
    if tp1_hit:
        fill = tp1_p + (SLIPPAGE if pos_dir == -1 else -SLIPPAGE)
        pnl_pts = (fill - entry_price) * pos_dir
        pnl_aud = pnl_pts * QTY - COMMISSION * 2

        trades.append({
            "entry_bar": entry_bar,
            "exit_bar": i,
            "entry_price": entry_price,
            "exit_price": fill,
            "direction": "L" if pos_dir == 1 else "S",
            "pnl_pts": pnl_pts,
            "pnl_aud": pnl_aud,
            "exit_type": "TP1",
            "flip_num": flip_num_at_entry,
            "date": cur_date,
        })
        equity += pnl_aud
        tp1_done = True
        entry_price = fill # move reference to TP1 for BE
        sl_p = fill # BE
        trail_armed = False
        trail_ref = fill

# 4. EOD close
if sess_end and pos_dir != 0:
    fill = c - (SLIPPAGE if pos_dir == 1 else -SLIPPAGE)
    pnl_pts = (fill - entry_price) * pos_dir
    pnl_aud = pnl_pts * QTY - COMMISSION * 2

    trades.append({
        "entry_bar": entry_bar,
        "exit_bar": i,
        "entry_price": entry_price,
        "exit_price": fill,
        "direction": "L" if pos_dir == 1 else "S",
        "pnl_pts": pnl_pts,
        "pnl_aud": pnl_aud,
        "exit_type": "EOD",
        "flip_num": flip_num_at_entry,
        "date": cur_date,
    })
    equity += pnl_aud
    pos_dir = 0

```



```

pct_weeks_pos = (weekly > 0).mean() * 100

print("=" * 60)
print("AU200 UT Bot Session v1 – Backtest Results")
print("BUG-019 FIXED | Slippage=1pt | Commission=$1/side")
print(f"Period: {tdf.date.min().date()} → {tdf.date.max().date()}")
print("=" * 60)
print(f"Trades:           {n_trades}")
print(f"Win Rate:          {wr*100:.1f}%")
print(f"Avg Win:           {avg_win:.1f} pts")
print(f"Avg Loss:           {avg_loss:.1f} pts")
print(f"Profit Factor:       {pf:.2f}")
print(f"Net Pts:             {total_pts:.1f}")
print(f"Net AUD:             ${total_aud:,.0f}")
print(f"Max Drawdown:        {max_dd:.1f}%")
print(f"")
print(f"Weekly stats (pts):")
print(f"  Median:             {weekly.median():.1f}")
print(f"  Mean:               {weekly.mean():.1f}")
print(f"  Std:                {weekly.std():.1f}")
print(f"  % weeks +ve:        {pct_weeks_pos:.0f}%")
print(f"  Best week:          {weekly.max():.1f}")
print(f"  Worst week:         {weekly.min():.1f}")
print(f"")
print("Year-by-year:")
for yr, grp in tdf.groupby("year"):
    yr_pts = grp.pnl_pts.sum()
    yr_wr = (grp.pnl_pts > 0).mean() * 100
    print(f"  {yr}: {grp.shape[0]:4d} trades | WR {yr_wr:.0f}% | {yr_pts:+.0f} pts")

print()
print("Exit type breakdown:")
print(tdf.groupby("exit_type").agg(n=("pnl_pts", "count"), pts=("pnl_pts", "sum"), wr=("pnl_pts", lambda x: (x>0).mea

print()
print(f"MINTICK confirmed from data: 0.1 pt")
print(f"Slippage used: {SLIPPAGE} pt/side ({SLIPPAGE/0.1:.0f} ticks)")

```

DOCUMENT END

ONE-LINE CURRENT STATUS

AU200: NO ROBUST CHAMPION. PF 0.83 is the best verified result (BUG-019-corrected Python backtest, 6 years of real data). No AU200 row in RESULTS_LEDGER. The 15:00 AEST session has raw edge (t=3.77) but no strategy has captured it profitably.

Sources confirmed included:

Source	Included
trader_playbooks/AU200_KNOWLEDGE_BASE.md	YES — full content
research/au200_utbot_session_v1.pine	YES — full code (Section 8)
research/au200_utbot_backtest.py	YES — full code (Section 9)
MEMORY.md (AU200-relevant sections)	YES
BUG_REGISTRY.md (BUG-019 through DST bug)	YES — Section 6
Session chat: au200_base_flip_v1.pine spec	YES — Section 2.1
Session chat: au200_tbt_flip_backtest.pine spec	YES — Section 2.2
Session chat: nine-row screenshot table	YES — Section 2.7
Python edge scan results (hour-by-hour, gap, ORB, momentum)	YES — Sections 2.5/3.1
RESULTS_LEDGER.md (AU200 rows)	NONE EXIST